

Tandem Q

Tandem's two RMS soft-knee compressors in series, with a passive-style program EQ on the same panel.



Overview

Tandem Q is Tandem with a program EQ added below it. The top unit is two classic RMS / VCA compressors one after the other: each stage does a little of the work, which is the old studio trick for heavy compression that still sounds open. The bottom strip is a passive-style program EQ with LOW, LO MID, HI MID and HIGH sections, running through transformer stages.

Both compressor stages measure loudness (RMS) rather than peaks, so they react the way your ears do. There are no attack and release knobs - the timing follows the music: the further the signal goes over the threshold, the faster the stage reacts (about 15 ms for 10 dB over, 5 ms for 30 dB), and it recovers at a steady 120 dB per second.

ORDER chooses whether the EQ comes before or after the compressors. The unit the signal meets first is shown on top.

Reach for it for

- Vocals that need levelling and tone-shaping in one place
- Drum and mix buses that want glue plus a broad EQ lift
- Bass guitar: even level, then low-end weight from the LOW section
- Ducking a music bed under a voice (EXT KEY)
- Creative pumping and reverse-dynamics effects (negative ratios)

Controls

Stage 1 and Stage 2

The two stages have the same main controls. The signal goes through Stage 1, then Stage 2.

IN	Switches the stage in or out. With a stage out, the audio passes straight through it.
THRESHOLD (red cap)	The level where the stage starts working, from -60 to 0 dB. Turn it down (left) for more compression. Because the detector measures loudness, the threshold relates to the average level rather than the peaks.
RATIO (yellow cap)	How strongly the stage compresses once the level is over the threshold. 1:1 does nothing; 2:1 and 4:1 are moderate; INF:1 (marked in red) is limiting - nothing gets louder. Past INF the ratio becomes negative (-2:1, -1:1): the louder the input, the quieter the output. That reverses the dynamics, for pumping and swelling effects.
OUTPUT (blue cap)	The stage's make-up gain, from -20 to +20 dB. Use it to bring the level back up after compression.
BELOW / KNEE / ABOVE	Three lights showing where the level sits: green below the threshold, amber in the soft knee, red above it.
GR meter	The row of 12 lights under the knobs shows the gain reduction of that stage, from 1 to 30 dB.
HARD	Off (default): a soft knee - the ratio eases in gradually over 10 dB around the threshold, which is smoother and more

transparent. On: the full ratio starts right at the threshold, for a firmer, more obvious effect.

Stage 1 only

DECOUPLE	Off: both sides of the stereo signal get exactly the same gain, so the stereo image stays still. On: each side mostly follows its own level (80% its own, 20% shared), like unlinking a bus compressor. The sound becomes wider and livelier, but the image can move a little.
EXT KEY	Stage 1 listens to the key input (plug-in pins 3 and 4) instead of the audio it is compressing. Use it for ducking: route a voice or kick to pins 3/4 and the music dips whenever it plays. Stage 2 always listens to Stage 1's output.
KEY IN	Lights when a signal arrives on the key input - a quick check that your routing works.

Stage 2 only

GRIT	Drives Stage 2's VCA into gentle 2nd and 3rd harmonic distortion. The harder Stage 2 works, the grittier it gets, so it adds attitude on the loud parts and stays clean on the quiet ones.
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Master

Transfer curve	Shows the input-to-output curve of both stages together. The white dot rides on it with the live level, so you can see exactly where the music sits.
SC HPF	A high-pass filter on what both stages listen to (fully left = off, up to 400 Hz). Raise it so the bass doesn't make the whole mix pump; the bass itself isn't filtered.
DRIFT	A slow random wander in the program EQ, like old components warming up, 0 - 100% (50% is the standard amount, 100% twice that, 0% none). It nudges each side's

	EQ frequencies and gains independently and very slightly, so the EQ breathes a little. It does not move the compressors.
MIX	Parallel compression: blends the dry signal back in. 100% is fully compressed; lower settings keep the transients of the dry signal under the compressed sound.
IN / OUT meters, GR	Input and output level, and the combined gain reduction of both stages in dB.

Program EQ

The strip below the compressors. Each frequency knob clicks through fixed steps, as on passive hardware.

IN	Switches the EQ curves in or out. The change glides rather than jumps.
LOW: FREQ	The low shelf frequency: 20, 30, 60, 100 or 150 Hz.
LOW: BOOST	Lifts the lows, 0 - 10. The boost is a resonant shelf with a little bump just above the frequency.
LOW: ATTEN	Cuts the lows, 0 - 10, with a very broad, gentle shelf. Using BOOST and ATTEN together is the old passive-EQ trick: the deepest lows still lift while a dip forms just above them, tightening the low mids.
LO MID: FREQ	150, 220, 330, 470, 680 Hz, 1 kHz or 1.5 kHz.
LO MID: GAIN	Boost or cut, -12 to +12 dB. The band gets narrower as you turn it further from 0.
LO MID: NARROW	Off: a wide, musical bell. On: a narrower bell for more focused changes.
HI MID: FREQ	1, 1.5, 2.2, 3.3, 4.7, 6.8 or 10 kHz.
HI MID: GAIN / NARROW	As LO MID: -12 to +12 dB, and NARROW for a tighter bell.

HIGH: FREQ	The centre of the high boost: 3, 4, 5, 8, 10, 12 or 16 kHz.
HIGH: BOOST	Lifts the highs around that frequency, 0 - 10.
HIGH: WIDTH	The width of the high boost, 0 - 10: low settings are sharp and focused, high settings broad and airy. Starts at 5.
HIGH: ATT FREQ	Where the high cut starts: 5, 10 or 20 kHz.
HIGH: ATTEN	Cuts the highs above ATT FREQ with a gentle shelf, 0 - 10. Combine with BOOST to lift presence while softening the very top.
OUT	The EQ's output level, -12 to +12 dB.

The header

ORDER	COMPRESSOR > EQ or EQ > COMPRESSOR: click to swap. The panel redraws with the unit the signal meets first on top.
PRESETS	Opens the list of factory presets; the ◀ / ▶ keys step through them and the display shows the one in use. Choosing one sets the two compressor stages, SC HPF and MIX; the EQ and DRIFT are left as they are. The last entry resets the compressor and DRIFT to their defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Input guard

There is no input or output fold-out on Tandem Q. An input guard runs in front of everything and is on by default: it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. It has no key on the panel; it appears as Input guard in REAPER's parameter list.

Presets

Gentle Glue	Light two-stage smoothing for a mix or group bus.
Vocal Leveller	Stage 1 evens the performance, Stage 2 catches the peaks; a little SC HPF.
Vocal Upfront	Heavier vocal control with a hard knee and GRIT on Stage 2.
Drum Bus Stack	Stage 1 decoupled for a lively image, GRIT on Stage 2, 70% mix.
Parallel Crush	Stage 2 limiting hard with GRIT, blended in at 35%.
Bass Even	Steady bass level with a little GRIT.
Acoustic Guitar	Gentle, with Stage 1 decoupled for width.
Peak Catcher	Light levelling, then a hard-knee limit at -6 dB.
Negative Pump	Stage 2 past INF for rhythmic pumping, 60% mix.
Reverse Swell	Stage 1 out, Stage 2 at -1:1: loud hits dip and the tails swell up; 50% mix.
Ducker (ext key)	Stage 1 ducks from the key input - route the voice or kick to pins 3/4.
Reset all to defaults	The compressor controls and DRIFT back to their starting values (the EQ is left as it is).

Using the key input (EXT KEY)

1. On the track with Tandem Q, open ROUTE and set the track to 4 channels.
2. On the track you want to duck from (e.g. the voice), add a send to Tandem Q's track and set its destination to channels 3/4.
3. Switch EXT KEY on. KEY IN lights when the key signal arrives; lower Stage 1's THRESHOLD until the music dips by the amount you want.

Tips

- EQ > COMPRESSOR lets the compressor react to your EQ: boosting the lows makes it work harder on the bass. COMPRESSOR > EQ keeps the EQ's tone moves out of the compressor's way.
- Start with Stage 2's ratio at INF and its threshold just under your peaks: it becomes a safety limiter after Stage 1's gentle levelling.
- If the low end makes the whole mix duck, raise SC HPF to 80-120 Hz.
- On a vocal, try a little HIGH BOOST at 10 or 12 kHz with WIDTH up for air, and a touch of LO MID cut at 220 or 330 Hz to clear mud.

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